CH4

子二甲 U13157005 吳翊弘

File Edit Selection View Go Run ... BlinkAssignment

EXPLORER

BLINKASSIGNMENT
- .vscode
- build
- port
- src
  - Agent.cpp
  - Agent.h
  - BlinkAgent.cpp
  - BlinkAgent.h
  - BlinkHeavy.cpp
  - BlinkHeavy.h
  - BlinkWorker.cpp
  - BlinkWorker.h
  - CMakeLists.txt
  - main.cpp 8
- .gitignore
- CMakeLists.txt
- FreeRTOS_Kernel_import.cmake
- pico_sdk_import.cmake

OUTLINE

TIMELINE

main.cpp 8 | CMakeLists.txt

src > main.cpp > blink_leds_reverse()

```cpp
/***
// U13157005 吳翊弘

#include "pico/stdlib.h"
#include "FreeRTOS.h"
#include "task.h"
#include <stdio.h>


#define NUM_LEDS 29
uint8_t led_pins[NUM_LEDS] = {
    2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 2
};


#define LED_BLINK_PRIORITY tskIDLE_PRIORITY + 1

void blink_leds_forward() {
    for (int i = 0; i < NUM_LEDS; i++) {
        gpio_put(led_pins[i], 1);
        vTaskDelay(500 / portTICK_PERIOD_MS);
        gpio_put(led_pins[i], 0);
    }
}


void blink_leds_reverse() {
    for (int i = NUM_LEDS - 1; i >= 0; i--) {
        gpio_put(led_pins[i], 1);
        vTaskDelay(500 / portTICK_PERIOD_MS);
```

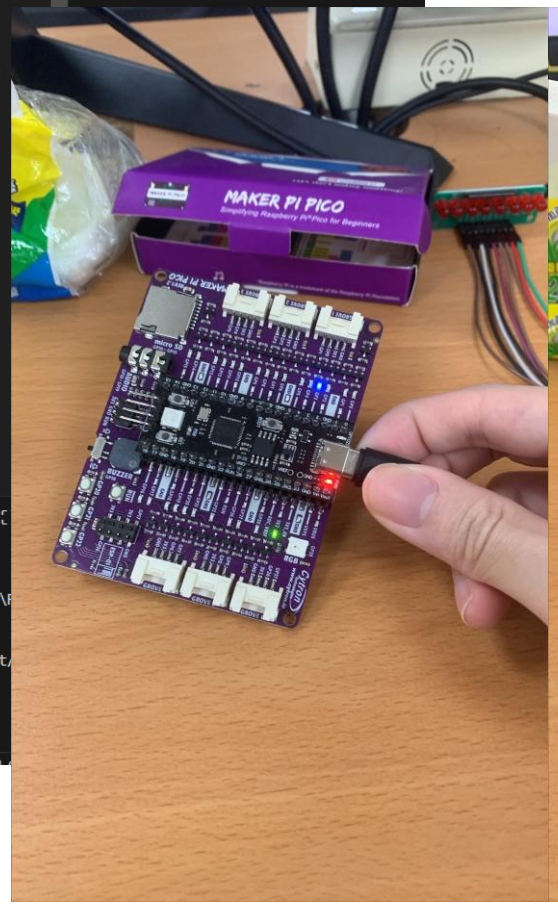
PROBLEMS 8 | OUTPUT | DEBUG CONSOLE | TERMINAL | Compile Project - Task

```
* Terminal will be reused by tasks, press any key to close it.

* Executing task: C:\Users\LAB216/.pico-sdk/ninja/v1.12.1/ninja.exe -C C:\Users\LAB216\Downloads\111\
coFreeRTOSCourse8\4-Tasks\BlinkAssignment/build

ninja: Entering directory `C:\Users\LAB216\Downloads\111\RPIPicoFreeRTOSCourse8\4-Tasks\BlinkAssignment/
d'
[2/2] Linking CXX executable src\Blink.elf
* Terminal will be reused by tasks, press any key to close it.
```

CHAT

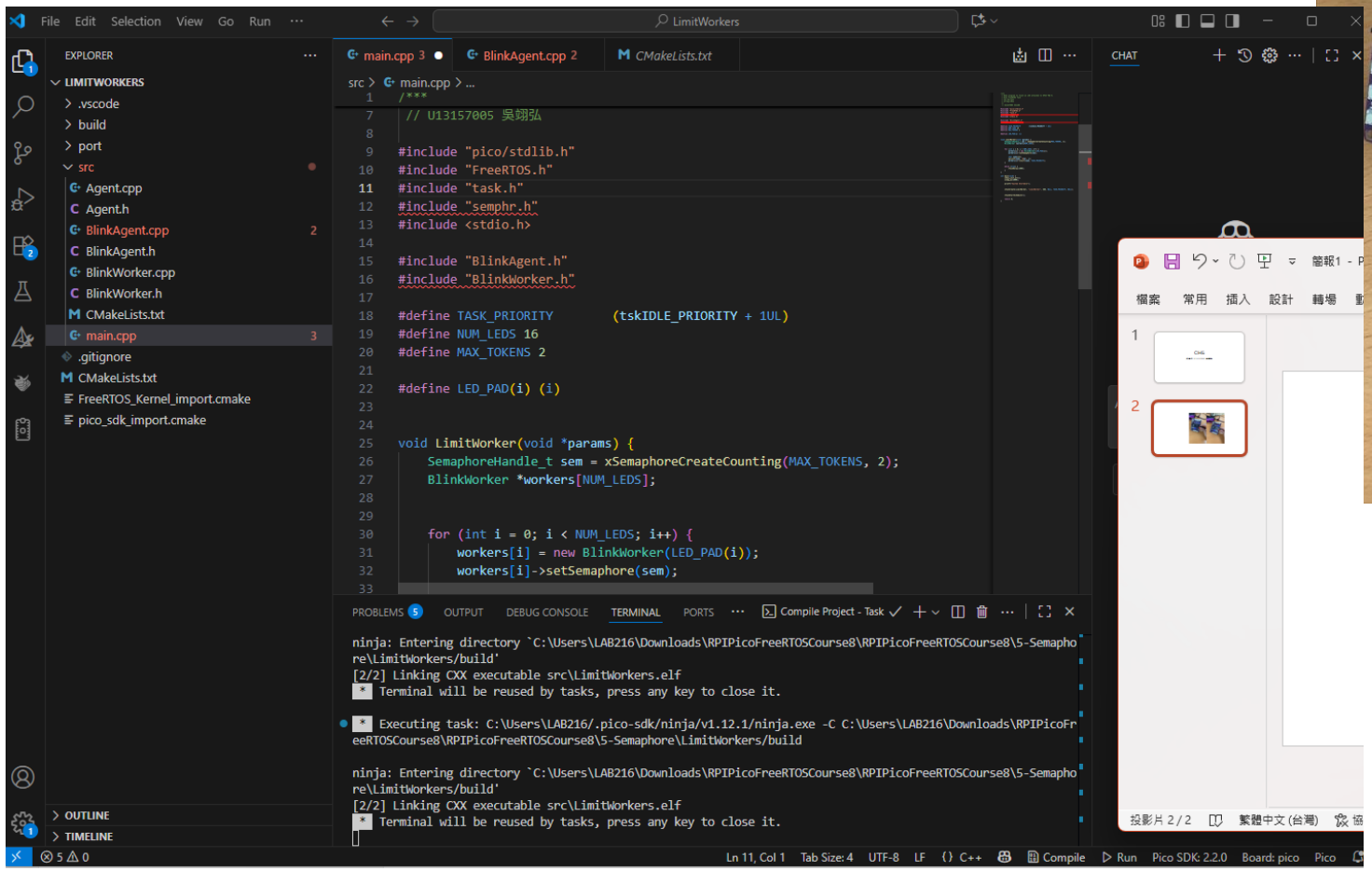


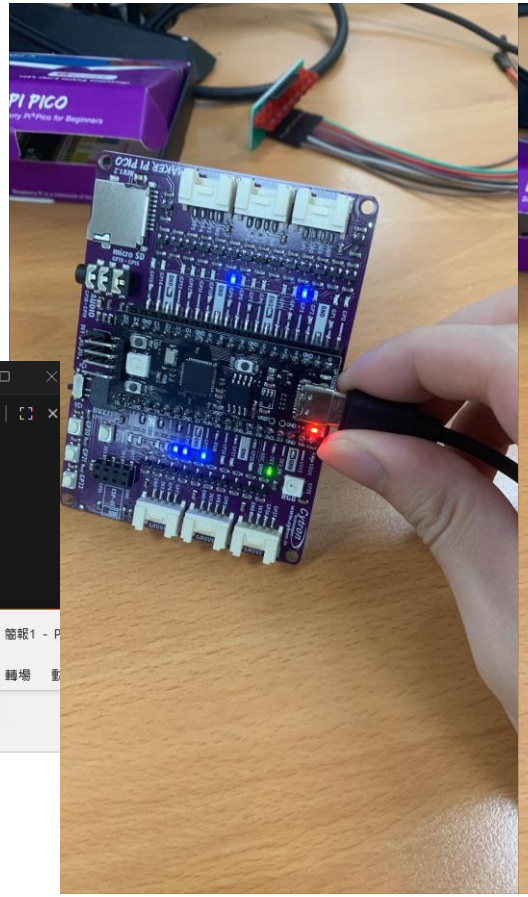
READ ME

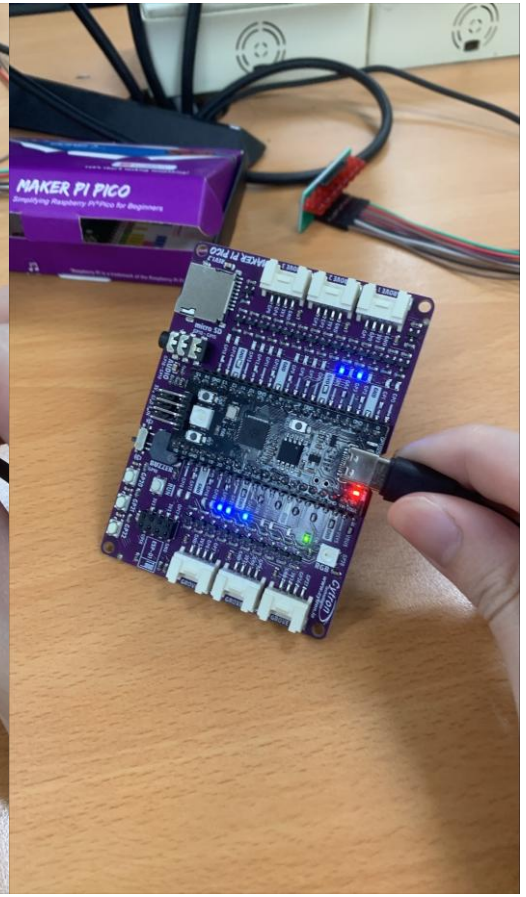
控制 29 顆 LED 按照正序1~29和反向29~1閃爍。每顆 LED 會亮起 500 毫秒，然後熄滅。使用者只需將 29 顆 LED 連接到 ，編譯並燒錄程式到 RP2040 開發板，LED 會依次閃爍，實現簡單且高效的多任務控制。

CH5

子二甲 U13157005 吳翊弘


File Edit Selection View Go Run
LimitWorkers
EXPLORER
LIMITWORKERS
.vscode
build
port
src
Agent.cpp
Agent.h
BlinkAgent.cpp 2
BlinkAgent.h
BlinkWorker.cpp
BlinkWorker.h
CMakeLists.txt
main.cpp 3
.gitignore
CMakeLists.txt
FreeRTOS_Kernel_import.cmake
pico_sdk_import.cmake
OUTLINE
TIMELINE
main.cpp 3
BlinkAgent.cpp 2
CMakeLists.txt
src > main.cpp > ...
/***
// U13157005 吳翊弘
#include "pico/stdlib.h"
#include "FreeRTOS.h"
#include "task.h"
#include "semphr.h"
#include <stdio.h>
#include "BlinkAgent.h"
#include "BlinkWorker.h"
#define TASK_PRIORITY (tskIDLE_PRIORITY + 1UL)
#define NUM_LEDS 16
#define MAX_TOKENS 2
#define LED_PAD(i) (i)
void LimitWorker(void *params) {
SemaphoreHandle_t sem = xSemaphoreCreateCounting(MAX_TOKENS, 2);
BlinkWorker *workers[NUM_LEDS];
for (int i = 0; i < NUM_LEDS; i++) {
workers[i] = new BlinkWorker(LED_PAD(i));
workers[i]->setSemaphore(sem);
PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Compile Project - Task
ninja: Entering directory `C:\Users\LAB216\Downloads\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\5-Semaphore\LimitWorkers/build'
[2/2] Linking CXX executable src\LimitWorkers.elf
* Terminal will be reused by tasks, press any key to close it.
* Executing task: C:\Users\LAB216/.pico-sdk/ninja/v1.12.1/ninja.exe -C C:\Users\LAB216\Downloads\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\5-Semaphore\LimitWorkers/build
ninja: Entering directory `C:\Users\LAB216\Downloads\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\5-Semaphore\LimitWorkers/build'
[2/2] Linking CXX executable src\LimitWorkers.elf
* Terminal will be reused by tasks, press any key to close it.
Ln 11, Col 1 Tab Size: 4 UTF-8 LF {} C++ Compile Run Pico SDK: 2.2.0 Board: pico Pico
CHAT
檔案 常用 插入 設計 轉場
投影片 2 / 2 繁體中文 (台灣)


PI PICO


MAKER PI PICO

READ ME

- 觀察到 LED 閃爍的順序有隨機性，這是由於每次只有 2 顆 LED 任務同時執行，當兩顆 LED 任務完成後，其他 LED 任務才會開始閃爍。隨著時間的推移，閃爍的順序會變化。

- 在 2 顆 LED 同時閃爍 的情況下，其他 LED 任務需要等待信號量釋放後才能執行。

將 2 個信號量令牌 限制每次最多只有 2 顆 LED 同時閃爍 時，LED 閃爍的順序隨著時間和任務調度變得隨機。